

Week 4 · Heading Control

★ Reference material — read this first

The five courses below are **taught by the instructor of this course** and are the assumed background for it. They run from the fundamentals down to the graduate material, so start wherever the gap is. Every frame, symbol and derivation used here is developed in them at length; anyone whose prerequisites are thin should work through them first, then return.

#	Course	Level	Lang.	Video	Slides and code
1	Control Engineering (제어공학) — the foundation: transfer functions, feedback, stability, root locus, PID	undergraduate	KO	playlist	drive
2	Control System Design (제어시스템설계) — design rather than analysis: specifications, loop shaping, discrete implementation	undergraduate	KO	playlist	drive
3	Advanced Control Engineering (제어공학특론) — reference frames, the 6-DOF equation of motion, rotation matrices and Euler angles, linearisation and trim, vehicle control design	graduate	EN	playlist	drive
4	Sensor Signal Processing and Fusion (센서신호처리 및 융합) — sensor models, noise, estimation and multi-sensor fusion	graduate	EN	playlist	drive
5	Capstone Design (캡스톤디자인) — a vehicle project carried end to end	undergraduate	KO	playlist	drive

Not fluent in MATLAB or Simulink yet? Do these before Part 2. Every laboratory in this course is Simulink, and the Onramp courses are free and take a few hours each.

Tool	Where to start
MATLAB	MATLAB Onramp · Core MATLAB Skills
Simulink	Simulink Onramp · instructor's Simulink lectures part 1 · part 2

🔧 Getting the course files, and keeping them current (Windows)

The notes, models and scripts are kept in one Git repository that is updated through the semester. Clone it once; before every class, pull. A pull downloads only what has changed since the last one.

When	Where to run it	Command
once	PowerShell — installs Git for Windows	<code>winget install --id Git.Git -e</code>
once	the folder that will hold the course, e.g. <code>Documents</code>	<code>git clone https://github.com/wkyouncnu/Sensor-Signal-Processing-and-Fusion.git</code>
before every class	inside the cloned folder <code>Sensor-Signal-Processing-and-Fusion</code>	<code>git pull</code>

- `git pull` prints `Already up to date.` when nothing has changed, and otherwise lists the files it updated.
- The repository is private. When Git asks, sign in with the GitHub account the instructor has given access.
- Experiment on copies, not on the cloned files: copy a week's `WXX_simulink` folder elsewhere first, and a pull can never collide with local edits. If it already has, `git stash`, then `git pull`, then `git stash pop` sets the edits aside, updates, and puts them back.
- The MSS toolbox is not part of the repository. The weeks that simulate the Otter need it at `Tools\MSS` inside the cloned folder.

- **Course:** USV Guidance, Navigation and Control (Graduate)
- **Department:** Autonomous Vehicle System Engineering, Chungnam National University
- **This week:** ① the Week 2 controller around the Otter's heading, tuned from the Scope with no model of the vessel ② why P alone reaches the command here and D damps again — the opposite of Week 3 ③ the crab angle: heading is not course ④ the moment limit, the back-calculation gain, and the wrap at $\pm 180^\circ$

★ Prerequisites from the previous week

- From Weeks 2 and 3: the five time-domain metrics, what P, I and D do, the filtered derivative, back-calculation anti-windup and the tuning order. Nothing new is added to the controller this week.
- From Week 3 §3-3: the same derivative term that damps a position loop made the speed loop worse. A term's effect is read from the axis, not assumed.
- From Week 1: the two propellers sit $y_{\text{pont}} = 0.395$ m either side of the centreline, so a difference in their thrust turns the vessel.

Learning Outcomes

Upon completion of this week, the learner is able to:

1. Characterise the yaw axis from the outside with one step of yaw moment, and explain from the result why P alone leaves no heading error.
2. Predict, from Weeks 2 and 3 and one fact about the axis, how P, I and D each change the heading response, and confirm it on the Scope.
3. Distinguish heading from course, and compute the crab angle from the body velocities.
4. Remove the heading error left by a weak propeller with the integral, and choose the back-calculation gain from a measured trade-off.
5. Explain why a heading error must be wrapped into $(-180^\circ, 180^\circ]$, and measure the cost of not wrapping it.

Prerequisites and Setup

Item	Requirement
MATLAB	R2024b, with Simulink
MSS	<code>Tools/MSS</code> , found by <code>W04_0_setup</code> through <code>mss_path</code>
Course folder	<code>lectures/W04_simulink</code>
Models	one per example — <code>W04_C_open_loop</code> , <code>W04_D_P</code> , <code>W04_E_PD</code> , <code>W04_F_PID</code> , <code>W04_G_wrap</code> , <code>W04_H_tuning</code> — all generated by <code>W04_1_build_heading</code> , never edited by hand
Time	lecture 40 min, laboratory 1 h 20 min

Part 1 · Principle and experiment, section by section

Every section states what is observed, says what follows from it, and then runs **the experiment that measures it**, on its own model. The numbers quoted in the text are printed by the run a few lines below it.

Part of a section	What it holds
What is observed	the effect itself, with the numbers it produces
What follows from it	numbered lines, one step each, resting on Weeks 2 and 3 and on one measurement of this axis
Experiment N-x	the model, the commands, the output actually obtained, the figure, and a table that reads every feature of the figure back to the numbered line that predicts it

4-0. Setting up (10 min)

```
cd lectures/W04_simulink
W04_0_setup
W04_1_build_heading
```

Expected output:

```
W04_0_setup
  thrust   X_ff = 60 N forward, yaw moment |N| <= 70.85 N m
  gains    Kp = 300   Ki = 20   Kd = 100   Nf = 20   Kb = 0.1
  command  psi_d = 0 -> 10 deg at t = 5 s

built W04_C_open_loop.slx (overlapping lines: 0)
built W04_D_P.slx         (overlapping lines: 0)
built W04_E_PD.slx        (overlapping lines: 0)
built W04_F_PID.slx        (overlapping lines: 0)
built W04_G_wrap.slx       (overlapping lines: 0)
built W04_H_tuning.slx     (overlapping lines: 0)
built W04_I_rate.slx       (overlapping lines: 0)
```

- `W04_0_setup.m` is the only file edited by hand. If it stops in `mss_path`, the MSS toolbox is not at `Tools\MSS`.
- The first line of `W04_0_setup` is `clear`. Run it again whenever a Scope does not match these notes.

4-1. The same controller, a new output

This section answers: what changes when the controller of Weeks 2 and 3 steers the vessel instead of speeding it up?

The controller does not change. The error, the three terms, the derivative filter, the limit and the back-calculation path are the blocks of Week 3, in the same places. Three things around them are new.

New block	What it does	Why
deg to rad	converts the command from degrees to radians	people command headings in degrees; the vessel model works in radians
ssa	wraps the error into $(-\pi, \pi]$: $ssa(e) = atan2(\sin e, \cos e)$	-170° is 20° past 170° , not 340° back; §4-6
allocation	a MATLAB Function block, commented line by line: turns the demanded yaw moment N into two propeller thrusts, then two shaft speeds	the Otter has no rudder; it turns by pushing one side harder

The yaw moment comes from a thrust difference. Both propellers push forward to give the surge force X_{ff} ; a yaw moment is added by pushing the port propeller harder and the starboard one less. Each propeller sits y_{pont} from the centreline, so a thrust difference $T_1 - T_2$ gives a moment $y_{pont}(T_1 - T_2)$, and the two thrusts are

$$T_1 = \frac{X_{ff}}{2} + \frac{N}{2y_{pont}} \text{ (port)}, \quad T_2 = \frac{X_{ff}}{2} - \frac{N}{2y_{pont}} \text{ (starboard)}$$

Symbol	Quantity	Value · source
X_{ff}	surge force from both propellers together	60 N, w04_0_setup.m
N	yaw moment demanded by the controller	N·m
y_{pont}	propeller arm, half the distance between the pontoons	0.395 m, otter_config
N_{max}	largest yaw moment available while X_{ff} is delivered	70.85 N·m: the port propeller reaches its 119.68 N maximum at $N = 2y_{pont}(119.68 - 30)$

- Each thrust is then turned into a shaft speed exactly as in Week 3, $n = \text{sign}(T)\sqrt{|T|/k}$.
- The moment limit block holds N inside ± 70.85 N·m. It plays the role the thrust limit played in Week 3, and it causes the windup of §4-5.

Experiment 4-1 · The Week 3 controller, steering instead of speeding up (10 min)

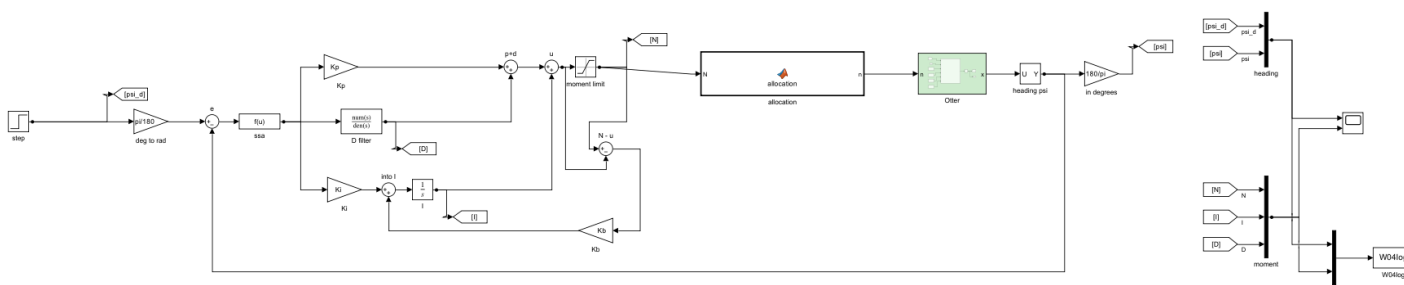
What it measures. Nothing yet: the canvas is matched to the three new blocks and to the thrust split of the formula above, so that each is recognised when it matters.

The canvas. w04_H_tuning — the fullest model of the week, read here and measured in Experiment 4-5. Everything between e and u is the Week 3 canvas unchanged.

Opening and running.

```
w04_0_setup
open_system('w04_H_tuning')           % Run - 위 칸 선수각, 아래 칸 모멘트 / heading above, moment below
```

In the law	On the canvas	Which changed since Week 3
the command in degrees	<code>step</code> → <code>deg to rad</code>	new: people command headings in degrees
$e = \text{ssa}(\psi_d - \psi)$	<code>e</code> → <code>ssa</code>	new: the wrap of §4-6
$K_p e + K_i \int e - K_d r$	<code>Kp</code> , <code>Ki</code> → <code>I</code> , <code>D filter</code>	the derivative now acts on the measured rate (§4-3 line 4)
the actuator's limit	<code>moment limit</code>	the quantity: ± 70.85 N·m instead of a force
T_1, T_2 and two shaft speeds	<code>allocation</code>	new: a moment out of two propellers



WEEK 4 SECTION IV - A BIG TURN: THE MOMENT LIMIT AND THE TUNING ORDER

$e = \text{wrap}(\psi_d - \psi)$ in rad, $N = K_p e + K_i \int e + K_d \dot{e}$ ($\psi = \text{deg}$)
 N is limited to $\pm M$ (max), what the propellers can give while pushing K_i , K_d ahead
 anti-windup I , $\dot{e} = K_i e + K_d (N - \dot{e})$, $K_d = 0$ switches it off

Change a gain in the Command Window (e.g. $K_p = 300$) and press Run.
 The Scope shows the heading in degrees (`psi`) and the yaw moment (`moment`).

In the figure	Meaning
<code>step</code> → <code>deg to rad</code> → <code>e</code> → <code>ssa</code>	the command in degrees, the error in radians, wrapped
<code>Kp</code> , <code>D filter</code> , <code>Ki</code> → <code>I</code> , <code>p+d</code> , <code>u</code>	the PID of Weeks 2 and 3, block for block
<code>moment limit</code> , <code>N - u</code> , <code>Kb</code>	the moment limit and the back-calculation path of Week 3
<code>allocation</code>	T_1 and T_2 of §4-1, the port propeller's efficiency <code>port_eff</code> (1 unless Experiment 4-3c lowers it), and the two shaft speeds; double-click it to read the commented code
<code>Otter</code> → <code>heading psi</code> → <code>in degrees</code>	the vessel, and the heading back in degrees for the Scope
<code>heading</code> , <code>moment</code> → Scope	top: ψ_d and ψ ; bottom: N , I and D

In class

- **Purpose** — show that the controller is still the Week 2 controller; the new parts are the degree conversion, the wrap, and the thrust split that makes a moment out of two propellers.
- **Point to** — open the `allocation` block behind the `moment limit`: one moment in, two thrusts out, one larger and one smaller, each turned into a shaft speed.
- **Ask** — "Why is there a `moment limit` at all, when each propeller has its own limit?" Because the demand must be limited where the controller can see it; otherwise the integral cannot know it was cut, and windup follows (§4-5).
- **Take away** — to steer a rudderless vessel, push one side harder.

4-2. What the vessel does, measured from outside

This section answers: what can one step of yaw moment reveal about the heading?

Model `w04_C_open_loop` has no controller. The vessel runs ahead on $X_{ff} = 60$ N, and at $t = 5$ s a constant yaw moment N is added. Measured by Experiment 4-2, below:

N [N·m]	final turn rate [deg/s]	turn rate per N·m	time to 63 % of the rate [s]	heading at 40 s [deg]
5	3.05	0.610	0.36	108.2
10	5.06	0.506	0.30	178.3
20	8.14	0.407	0.24	286.0

What follows from it, one line at a time.

- The heading never stops.** A constant moment gives a constant turn rate, and the heading grows for as long as the moment is applied: **108°**, **178°**, **286°** at 40 s. The heading is the running sum — the integral — of the turn rate.
- So the plant contains an integrator of its own.** In the language of Week 2 §2-4 it is **type 1** without any help from the controller, and by the final value theorem a proportional controller alone will therefore end **exactly** at the command. This is the opposite of Weeks 2 and 3, where P always left a gap.
- Holding a heading costs no moment.** Line 1 read backwards: zero moment gives zero turn rate, so a vessel already on its heading needs nothing to stay there. There is no drag and no spring on this axis to be held against — which is why line 2 comes out as it does.
- The turn rate answers quickly.** It is within 63 % of its final value in **0.36** s or less, so the slow part of a heading change is not the vessel's response but the **adding up** of the rate.
- The turn rate is not proportional to the moment.** Doubling the moment from 10 to 20 N·m raises the rate by only **61 %**: the yaw damping grows with the rate itself (Week 1: $N_r(1 + 10|r|)r$), so a quick turn meets more resistance. Unlike the surge speed of Week 3 this axis is visibly **nonlinear**, which is one more reason to tune by measurement rather than from a model.

Experiment 4-2 · The heading seen from outside (10 min)

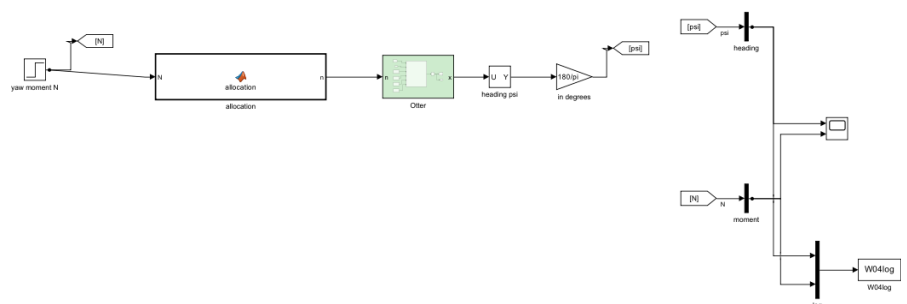
What it measures. Lines 1, 4 and 5: a heading that grows without end, a turn rate that settles within a third of a second, and a rate per N·m that falls as the moment grows.

The model. `w04_C_open_loop` — no controller. The vessel runs ahead on $X_{ff} = 60$ N and a constant yaw moment is added at $t = 5$ s.

Opening and running.

```
w04_0_setup
open_system('w04_C_open_loop')    % Run - 선수각이 멈추지 않는다 / the heading never stops
N_open = 20;                      % Run - 모멘트는 네 배, 회두율은 2.7 배뿐 / not proportional
w04_C_plant_from_outside          % 세 모멘트를 한 번에 / all three moments at once
```

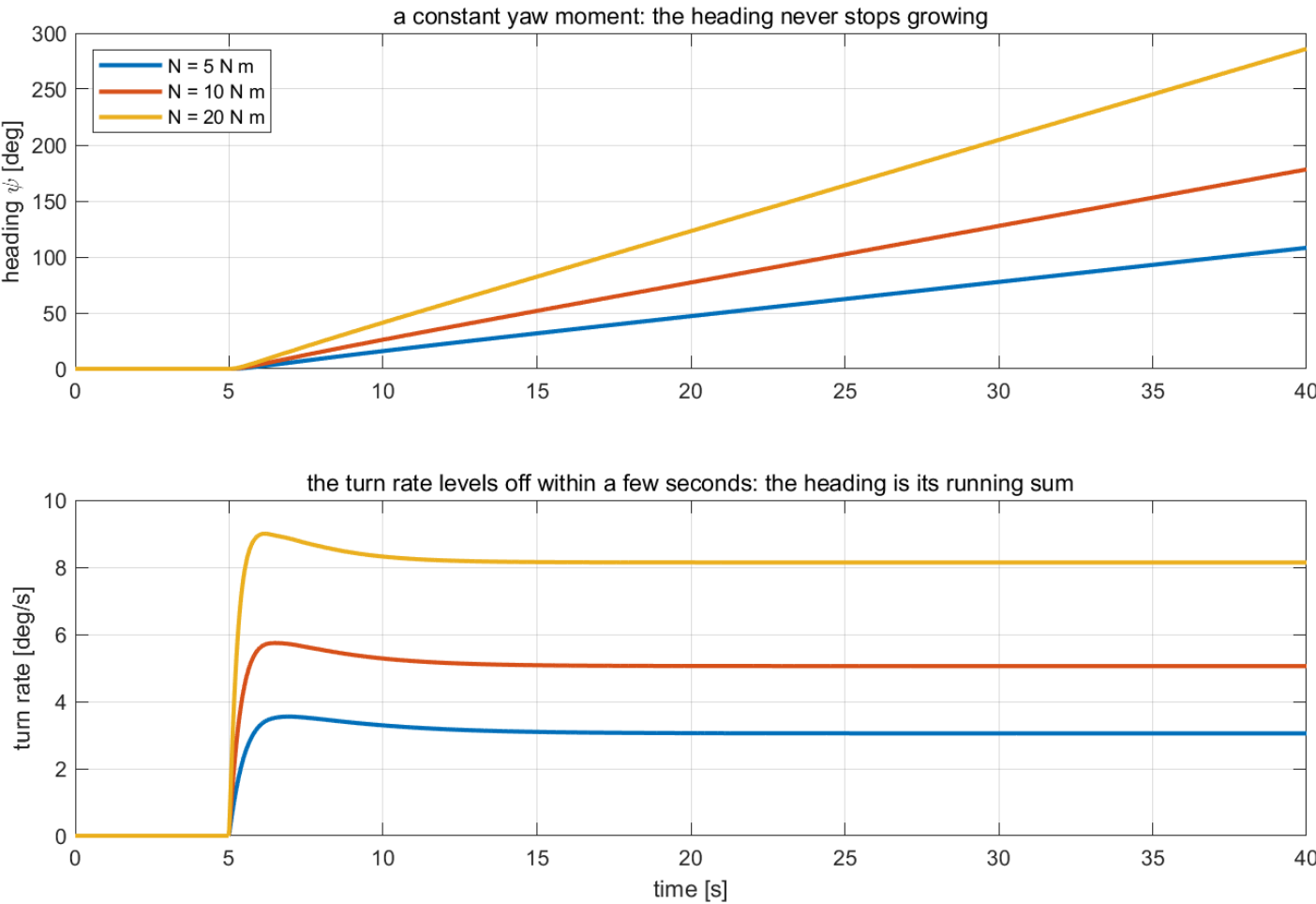
WEEK 4, SECTION C - THE PLANT, SEEN FROM OUTSIDE (no controller)
N, open reaction matrix of yaw moment from 1 × 1, stop, with X, 8 newtons ahead
Change N, open (5, 10, 20) and press Run: does the heading ever stop?



In the figure	Meaning
yaw moment N	a step of N_{open} N·m at $t = 5$ s
the thrust split, Otter , heading ψ , in degrees	the plant chain of §4-1
Scope	top: the heading ψ ; bottom: the moment N

Expected output:

W04 Experiment 4-2 the plant seen from outside (no controller, $X_{\text{ff}} = 60$ N ahead)				
N [N m]	final turn rate [deg/s]	rate per N m	time to 63 % of the rate [s]	heading at 40 s
[deg]				
5	3.05	0.610		0.36
108.2				
10	5.06	0.506		0.30
178.3				
20	8.14	0.407		0.24
286.0				



In the figure	Meaning
top	the heading after a step moment of 5, 10 and 20 N·m
bottom	the turn rate, the slope of the top curves

Reading the figure against the derivation.

Where to look	What is there	Which line predicts it
top panel, the curves are straight and never level off	108°, 178°, 286° at 40 s	line 1: a constant rate integrated is a heading that grows without end
bottom panel, each rate flat after a fraction of a second	3.05, 5.06, 8.14 deg/s	line 4: the rate is the fast part; 0.36 s to 63 %
top panel, the slope of each straight line	equals the flat value below it	line 1 again: the heading is the running sum of the rate
the three flat rates against 5, 10, 20 N·m	4× the moment gives only 2.7× the rate	line 5: the yaw damping grows with the rate
the bottom panel before $t = 5$ s	zero moment, zero rate, heading unchanged	line 3: holding a heading costs nothing

What the figure says

- The heading grows without end while the turn rate levels off within a few seconds. The heading is the running sum of the turn rate.

What to try	What to watch
N_{open} = 40; Run	twice the moment of the last row gives 12.82 deg/s against 8.14 — a factor 1.57 , not 2: line 5 bites harder the faster the vessel turns
N_{open} = 0; Run	the rate is 0.00 deg/s and the heading is still 0° at 40 s: line 3, holding a heading costs no moment

In class

- Purpose** — contrast with Week 3's open loop: there a constant force gave a constant speed; here a constant moment gives a heading that never stops.
- Point to** — the straight lines in the top panel; the flat rates in the bottom one.
- Ask** — "What moment does it take to hold a heading?" None: a vessel on a steady heading needs no yaw moment. That is why P alone will leave no error (Experiment 4-3a).
- Ask** — "Why does 20 N·m give less than twice the rate of 10 N·m?" The yaw damping grows with the turn rate.
- Take away** — the heading is an integral; the plant brings its own.

4-3. P, I and D on the heading

This section answers: what does each term do on this axis, and how does it compare with Weeks 2 and 3?

What is observed. With the same three terms again, on this axis:

- P alone **rings** — overshoot **5.8** → **15.9 %** as K_p rises from 100 to 1000 — and yet leaves **no error at all**: the final heading is **10.000°**.
- D damps it, as in Week 2 and unlike Week 3: overshoot **12.18** → **0.39 %**, settling **3.26** → **1.72** s.
- I is not needed to reach the command, and becomes necessary only when something pushes back — a port propeller at **70 %** leaves PD **0.80°** off.

The heading is therefore the third of three different answers from the same controller, and the two lines of §4-2 explain all of them.

What follows from it, one line at a time.

1. **The heading can overshoot, because it is an integral.** The vessel keeps turning while the moment is removed, so the heading passes the command before the rate reaches zero — the store that Week 3's speed axis did not have.
2. **P alone leaves no error**, by §4-2 lines 2 and 3: the plant already contains an integrator, and holding a heading costs no moment. At the command the error is zero, hence $N = 0$, and zero is exactly what is needed. Compare Week 3, where holding a speed cost a steady **116 N** and P therefore had to keep an error to produce it.
3. **D damps here.** The derivative of a heading error is a **turn rate**, and resisting a rate is damping, as resisting the mass's velocity was in Week 2. On the speed axis of Week 3 the same derivative was an acceleration, which is why it hurt there. The term has not changed; the axis has, for the third time.
4. **Most of the models of this week differentiate the error**, exactly as Weeks 2 and 3 did: the block **D filter** is $K_d N_f s / (s + N_f)$ fed by $ssa(\psi_d - \psi)$. That is the textbook form, and it is kept in **w04_D_P** ... **w04_H_tuning** so that its one weakness can be measured rather than described. **The form the rest of this course actually uses is the other one**, $-K_d r$, and the subsection "Why the derivative is better taken from the yaw rate" derives it, measures it in Experiment 4-3e, and shows what happens when its sign is wrong.
5. **I is needed only against a steady disturbance.** A propeller delivering **70 %** of its thrust makes a yaw moment that never goes away; PD can oppose it only from error, and settles **0.80°** off, holding **4.19 N·m**. The integral finds that moment by itself — **4.15 N·m** at $K_i = 20$ — which is Week 2 §2-8 line 5 once more, with a fouled propeller in place of a spring.

Term	Week 2, position	Week 3, speed	This week, heading	Measured
P	faster, rings, leaves an error	faster, never rings, leaves an error	faster, rings , and leaves no error	Experiment 4-3a: overshoot 5.8 → 15.9 % as K_p rises 100 → 1000 ; final heading 10.000°
D	damps	makes it worse	damps again	Experiment 4-3b: overshoot 12.18 → 0.39 % , settling 3.26 → 1.72 s as K_d rises 0 → 100 at $K_p = 300$
I	removes the error	removes the error	removes an error only when something pushes back : a weak propeller	Experiment 4-3c: 0.80° left by PD with the port propeller at 70 %; removed by $K_i = 20$

Experiment 4-3a · P only (10 min)

What it measures. Lines 1 and 2: a heading that ends exactly at the command whatever the gain, and an overshoot that grows with the gain because the plant integrates.

The model. **w04_D_P** — the loop of §4-1 with $K_i = K_d = 0$.

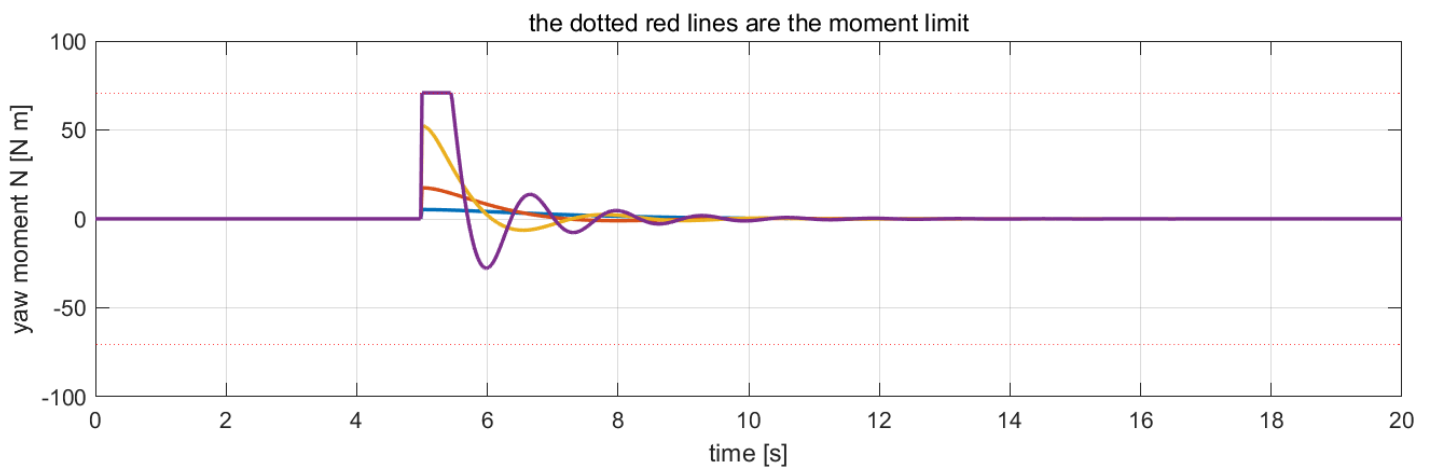
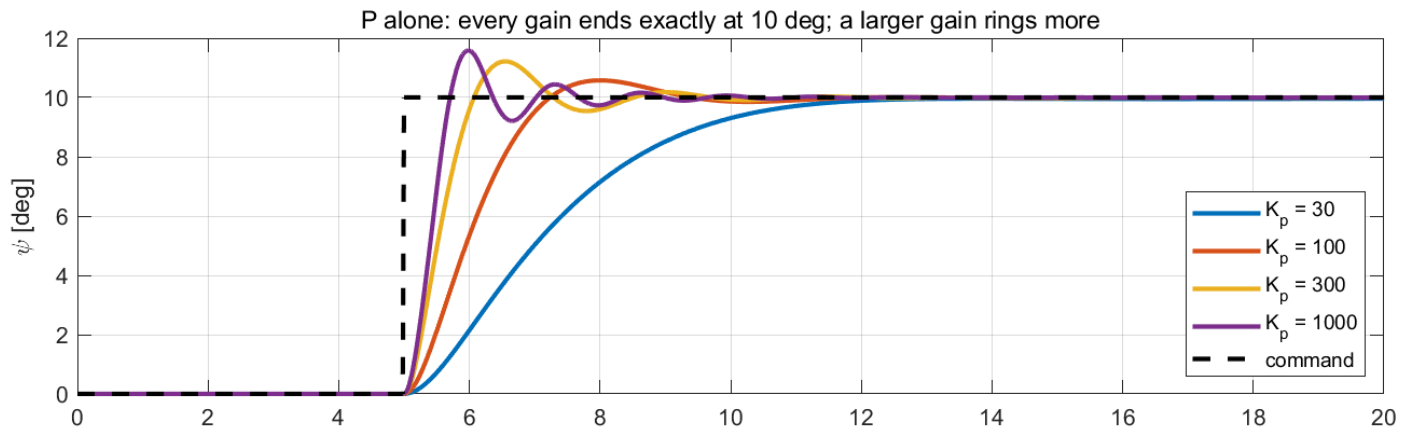
Opening and running.

```
w04_0_setup
open_system('w04_D_P')           % Run - 10.000 도에서 정확히 멈춘다 / it ends exactly at 10.000 deg
Kp = 1000;                       % Run - 빨라지고 더 넘친다 / faster, and it overshoots more
w04_D_proportional_only          % 네 게인을 한 번에 / all four gains at once
```

Expected output:

W04 Experiment 4-3a P only (psi_d = 10 deg at t = 5 s)

Kp	final psi [deg]	overshoot %	rise [s]	settle [s]	peak N [N m]
30	9.989	0.0	3.94	6.36	5.2
100	9.997	5.8	1.50	3.96	17.5
300	9.999	12.2	0.74	3.26	52.4
1000	10.000	15.9	0.48	3.12	70.8



In the figure	Meaning
top	the heading for $K_p = 30$ to 1000 and the dashed command
bottom	the yaw moment; the dotted red lines are ± 70.85 N·m

Reading the figure against the derivation.

Where to look	What is there	Which line predicts it
top panel, where every curve ends	10.000° , on the dashed command	line 2: the plant's own integrator, and no moment needed to stay
top panel, the peak above the command	0.0 → 15.9 % as K_p grows	line 1: the vessel is still turning when the moment is taken away
bottom panel, the moment at the command	back to zero in every run	line 2 again: zero error, zero moment, and that is enough
bottom panel, the negative moment after the peak	present in every ringing run	line 1: the moment must be reversed to stop the turn the vessel is still doing
bottom panel at $K_p = 1000$, the flat top	on the 70.85 N·m line	§4-1: the moment limit, which is what §4-5 must then manage

What the figure says

- Every gain ends at **10°**; a larger gain is faster and rings more. At $K_p = 1000$ the moment is on its limit and more gain buys little.

What to try	What to watch
<code>Kp = 30;</code> Run	slow but still exact: 9.989° with no overshoot at all. Line 2 does not depend on the size of K_p
<code>psi_step = 90; Kp = 300; T_final = 60;</code> Run	the moment sits on its limit for 4.24 s , and the overshoot is smaller, 0.91 % against 12.18 % at 10° : a saturated P loop simply turns at its largest rate, and there is no integrator to store anything. §4-5 adds one, and the same turn then overshoots by 15.6 %

In class

- **Purpose** — the two differences from Week 3 on one plot: no error left, and ringing.
- **Point to** — the final-heading column, all **10°** to within **0.011°**; the overshoot column growing with the gain.
- **Ask** — "Week 3 at $K_p = 800$ still missed by 0.133 m/s. Why is this different?" Holding a speed needs force against drag; holding a heading needs no moment.
- **Take away** — on an integrating axis P reaches the command; the question becomes how much it rings.

Experiment 4-3b · Add D (10 min)

What it measures. Lines 3 and 4: the derivative damping the ringing of Experiment 4-3a, and the size of what it demands at the instant of the step.

The model. `W04_E_PD` — the same loop with the `D filter` branch switched on, $K_d N_f s / (s + N_f)$ fed by the error. K_i is still zero. Where that derivative should come from is the subject of the subsection below.

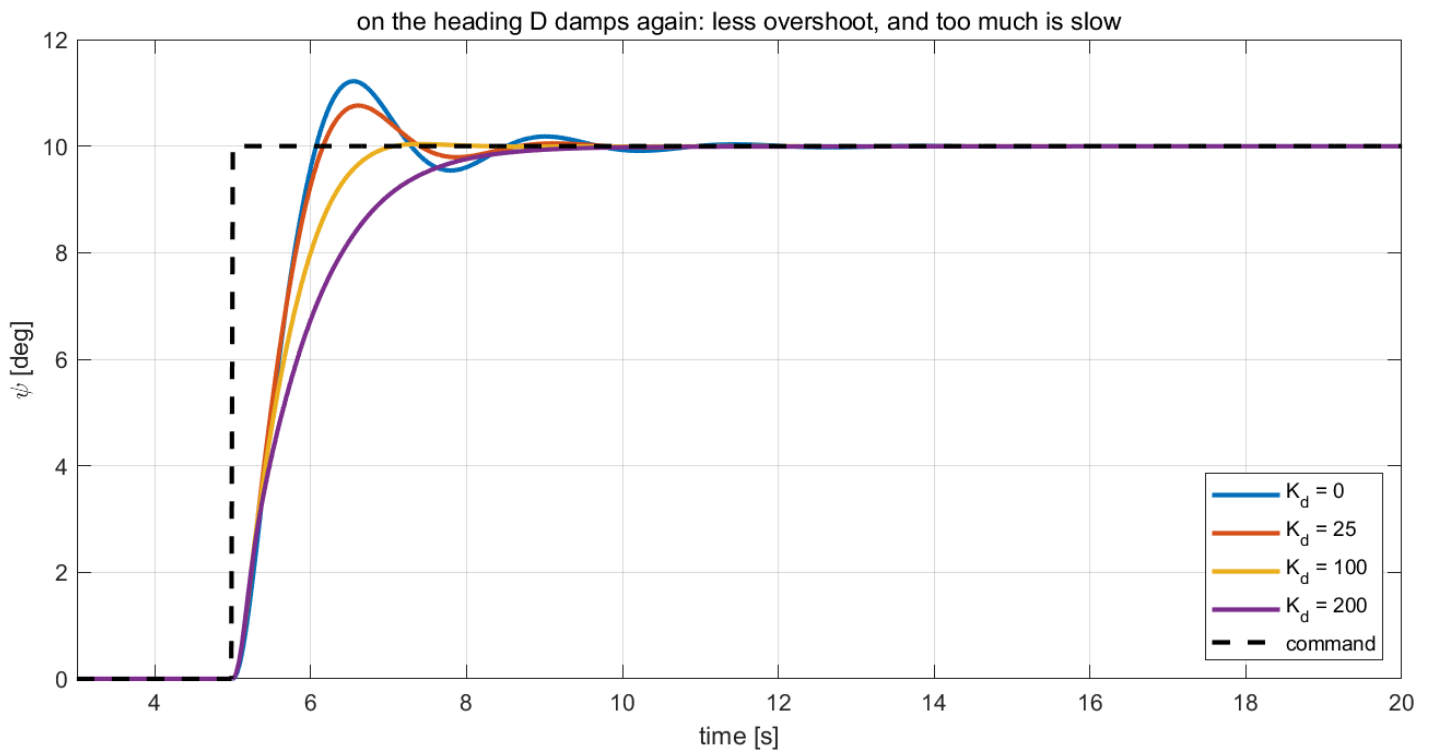
Opening and running.

<code>W04_0_setup</code>	
<code>open_system('W04_E_PD')</code>	% Run — $K_d = 100$: 거의 넘치지 않는다 / almost no overshoot
<code>Kd = 0;</code>	% Run — 4-3a 의 울림이 돌아온다 / the ringing of 4-3a is back
<code>Kd = 200;</code>	% Run — 넘침은 0, 대신 느리다 / no overshoot, but slow
<code>W04_E_derivative</code>	% 일곱 값을 한 번에 / all seven gains at once

Expected output:

W04 Experiment 4-3b P + D ($K_p = 300$, $\psi_d = 10$ deg)

K_d	overshoot %	rise [s]	settle [s]
0	12.18	0.74	3.26
10	10.23	0.76	3.20
25	7.61	0.80	2.98
50	4.10	0.88	2.22
100	0.39	1.12	1.72
150	0.00	1.44	2.42
200	0.00	1.78	3.14



In the figure	Meaning
traces	the heading for $K_d = 0, 25, 100, 200$ at $K_p = 300$

Reading the figure against the derivation.

Where to look	What is there	Which line predicts it
the peak of each trace	$12.18 \rightarrow 0.39\%$ as K_d grows to 100	line 3: a term against the turn rate is damping on this axis
the traces at $K_d = 150$ and 200	no overshoot, but rise 1.44 and 1.78 s	Week 2 §2-3: past $\zeta = 1$ more damping only costs speed
the settling column	falls to 1.72 s at $K_d = 100$, then rises	the two lines above pulling in opposite directions; the measurement picks $K_d = 100$
the moment at the instant of the step	a spike of 325.8 N·m, against a 70.85 N·m limit	line 4: the derivative of a step, $K_d N_f$ times the jump — the kick of Week 2 §2-10, and the reason for the subsection below

What the figure says

- The derivative removes the ringing; beyond $K_d = 100$ it only slows the turn.

What to try	What to watch
<code>Kd = 400;</code> Run	no overshoot at all, but the rise time is 3.24 s against 1.12 s at $K_d = 100$: the brake now outweighs the push
<code>Kp = 1000; Kd = 100;</code> Run	5.83 % of overshoot returns, where the same K_d left 0.39 % at $K_p = 300$ — K_d is read against the K_p it damps, never on its own

In class

- **Purpose** — the derivative on its home ground again, after it failed on the speed loop.
- **Point to** — the settling column: it falls to **1.72 s** at $K_d = 100$ and rises again — too much damping is slow (Week 2 §2-3, $\zeta > 1$).
- **Ask** — "The same block made Week 3 worse. What changed?" The axis: the derivative of a heading error is a turn rate, and resisting it is damping.
- **Take away** — use D where the output can overshoot, and stop raising it where the settling time turns up.

Experiment 4-3c · A weak port propeller, and the integral (10 min)

What it measures. Line 5: a disturbance that never goes away, the error PD keeps in order to oppose it, and the integral finding the same moment without an error.

The model. `W04_F_PID` — the full loop, with `port_eff` scaling the port propeller's thrust. At `port_eff = 1` nothing is wrong and the integral has nothing to do; at `0.7` the vessel is pulled off heading for as long as it runs.

Opening and running.

```
W04_0_setup
port_eff = 0.7; Ki = 0;           % 좌현 30 % 약함, 적분 없음 / a weak port propeller, no integral
open_system('W04_F_PID')         % Run - 9.200 도에서 멈춘다 / it stops 0.8 deg short
Ki = 20;                         % Run - 목표에 도달한다 / it reaches the command
W04_F_weak_propeller             % 두 스위치를 한 번에 / both sweeps at once
```

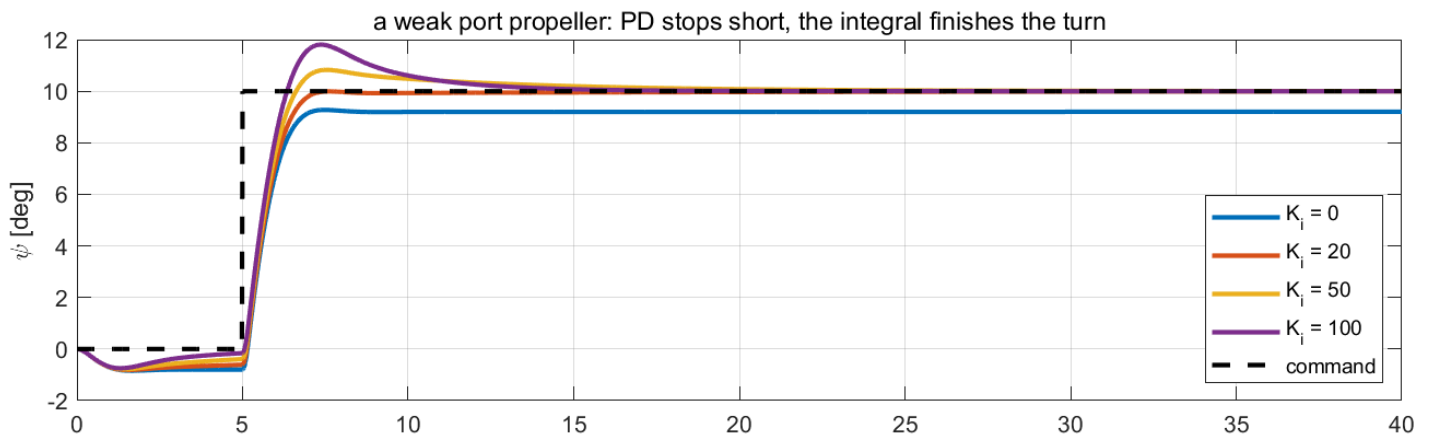
Expected output:

W04 Experiment 4-3c 1) PD only ($K_p = 300$, $K_d = 100$), port propeller at reduced efficiency

port_eff	final psi [deg]	error left [deg]	steady N [N m]
1	9.999	0.001	0.00
0.7	9.200	0.800	4.19
0.5	8.490	1.510	7.90

2) add the integral, `port_eff = 0.7`

Ki	final psi [deg]	overshoot %	settle [s]	I at 40 s [N m]
0	9.200	0.00	Inf	0.00
20	9.993	0.00	1.98	4.15
50	10.002	8.27	9.82	4.20
100	10.000	18.07	7.66	4.19



In the figure	Meaning
top	the heading with the port propeller at 70 %, for $K_i = 0$ to 100
bottom	the integral term; every curve ends at the same moment

Reading the figure against the derivation.

Where to look	What is there	Which line predicts it
top panel, the $K_i = 0$ curve	flat at 9.200°	line 5: PD can produce the needed 4.19 N·m only by keeping 0.80° of error
bottom panel, where every integral levels off	4.15 to 4.20 N·m, whatever K_i	Week 2 §2-8 line 5: an integrator stops where the error is zero, at the moment the fault demands
top panel, $K_i = 20$ against $K_i = 100$	0.00% against 18.07% of overshoot	too much integral is late moment, on an axis that already integrates twice over
the <code>port_eff</code> table, $1 \rightarrow 0.7 \rightarrow 0.5$	steady N of $0.00 \rightarrow 4.19 \rightarrow 7.90$ N·m	line 5: the worse the propeller, the larger the moment to be held, and the larger the error PD keeps

What the figure says

- PD stops 0.80° short, because holding the heading now needs 4.19 N·m that only an error can produce. The integral supplies it and the error goes; too much integral overshoots.

What to try	What to watch
<code>port_eff = 1; Ki = 20;</code> Run	with nothing to oppose, the integral settles at 0.22 N·m and the heading ends at 10.043° : an integral is insurance, not a gain to be raised for its own sake
<code>port_eff = 0.5; Ki = 20;</code> Run	the same K_i still reaches the command, 9.947° , now holding 7.64 N·m — the term finds whatever the fault costs

In class

- **Purpose** — show when the integral is needed on an axis that already integrates: when something pushes back.
- **Point to** — the steady N column: **4.19** N·m with PD, the same **4.15–4.20** N·m in the integral once $K_i > 0$.
- **Ask** — "Why does the weak propeller make a yaw moment at all?" Both propellers are asked for 30 N; the port one gives 21 N, so the starboard side pushes harder and the vessel turns to port.
- **Take away** — the integral finds whatever steady effort the plant demands, whether drag (Week 3) or a fault (here).

Why the derivative is better taken from the yaw rate

This subsection answers: the autopilot must brake the turn — why measure the **yaw rate** rather than differentiate the heading error?

The two forms. Both are called "the D term", and they are not the same expression:

$$\text{error form: } N = K_p e + K_d \dot{e}, \quad \text{rate form: } N = K_p e - K_d r, \quad e = \text{ssa}(\psi_d - \psi)$$

Why they are usually equal. Differentiating the error gives $\dot{e} = \dot{\psi}_d - \dot{\psi} = \dot{\psi}_d - r$. **Whenever the command is not moving**, $\dot{\psi}_d = 0$ and the two forms are identical, term for term. Everything the derivative does while the vessel settles on a fixed heading is the same in both.

Where they part, one line at a time.

1. They differ **only** through $\dot{\psi}_d$: the error form adds $K_d \dot{\psi}_d$, and nothing else.
2. A step command makes $\dot{\psi}_d$ an impulse. Filtered as §2-10 filters it, the term does not go to infinity but starts at $K_d N_f$ times the jump: $100 \times 20 \times 0.1745 \text{ rad} = 349 \text{ N·m}$ for the 10° step, against a moment limit of **70.85** N·m. **The rate form cannot do this**, because r is a velocity of a vessel with inertia and cannot jump.
3. That spike is spent on the actuator, not on the vessel: it is cut off by the limit within a few hundredths of a second, and §4-5 records the windup it causes even in a 10° turn.
4. **The seam makes it worse.** When a command crosses $\pm 180^\circ$, the wrapped error jumps by a full 2π — that is what `ssa` does, and §4-6 shows why it must. Differentiating that jump asks for $K_d N_f 2\pi = 12\,566 \text{ N·m}$, some 177 times the limit, for a turn of a few degrees. The rate form is untouched by it.
5. **The sensor decides it too.** A vessel carries a **gyroscope**, and r is what it reports directly. The error form has no measurement of $\dot{\psi}$: it must compute one by differentiating the heading, and a derivative multiplies every frequency by ω (§2-10) — so it amplifies heading noise, and then needs the filter N_f , whose phase lag costs some of the damping it was added for.
6. The price of the rate form is that it does not anticipate a **moving** command; a ramped command is followed with a small lag. A reference model that supplies ψ_d together with $\dot{\psi}_d$ removes even that, and is the subject of Week 8.

Why the sign must be negative. The rate term is written $-K_d r$ with $K_d > 0$, and the minus is not a convention — it is the whole of the term's purpose. Put the law into the yaw equation of §4-2 line 5:

$$M_{66} \dot{r} = N - \underbrace{|N_r|(1 + 10|r|)}_{\text{the hull's own damping}} r, \quad N = K_p e - K_d r$$

and with $e = \psi_d - \psi$ and $r = \dot{\psi}$ this is

$$M_{66} \ddot{\psi} + \left[\underbrace{|N_r|(1 + 10|r|)}_{\text{the hull's}} + \underbrace{K_d}_{\text{the controller's}} \right] \dot{\psi} + K_p \psi = K_p \psi_d$$

Symbol	Quantity	Value · source
M_{66}	yaw inertia, added mass included	42.65 kg·m ² · otter.m , Week 1 §1-9
N_r	linear yaw damping	−42.65 N·m per rad/s · $-M_{66}/T_{yaw}$, $T_{yaw} = 1$ s
K_d	the rate gain	100 N·m per rad/s

- K_d sits in the same bracket as the hull's own damping, and adds to it. That is what "damping" means here, and it is why the term is subtracted from the moment: a moment **against** the turn is a moment that removes energy from it.
- Flip the sign and the bracket becomes $|N_r|(1 + 10|r|) - K_d$. The controller now **subtracts** from the hull's damping, and with $K_d = 100$ against $|N_r| = 42.65$ the total is **negative** at low turn rates: the loop puts energy in on every swing. A brake wired backwards is an accelerator.
- It does not fly apart, and the reason is Week 1's nonlinear factor. The bracket returns to zero when $|N_r|(1 + 10|r|) = K_d$, that is at

$$|r| = \frac{1}{10} \left(\frac{K_d}{|N_r|} - 1 \right) = \frac{1}{10} \left(\frac{100}{42.65} - 1 \right) = 0.1345 \text{ rad/s} = 7.70 \text{ deg/s}$$

so the vessel grows its oscillation until the turn rate reaches about that value and then holds it — **a limit cycle, not a divergence**. Experiment 4-3e measures **7.23** deg/s.

What MSS does. Not one vessel in the toolbox differentiates the error. Three of them steer by the rate form, and they are the reference this course follows:

MSS file	The line	What it is
VESSELS/SIMclarke83.m	<code>M(3,3)*(Kp*ssa(psi_ref-eta(3)) - Kd*nu(3))</code>	a PD heading autopilot; $\nu(3)$ is r
VESSELS/SIMremus100.m	<code>delta_r = -Kp_psi*ssa(x(12)-psi_d) - Kd_psi*x(6)</code>	the same, with the error written the other way round; $x(6)$ is r
VESSELS/SIMrig.m	<code>tau3 = -R'*(Kp*eta3 + Ki*z3) - Kd*nu3</code>	3-DOF DP: P and I on the position error , D on the measured velocity ν

SIMrig is the one worth reading twice. The proportional and integral terms act on an error, and the damping term acts on a measured velocity — they are not three derivatives of one signal. That is exactly the structure Week 8 §8-3 builds and Week 9 assembles.

Two ways to write the same gain

Some texts — including the companion Capstone course — fold the minus into the gain and write $N = K_p e + K_{d,\psi} r$ with $K_{d,\psi} < 0$. This course keeps $K_d > 0$ and writes the minus in the law, so that every gain in every week is positive. They are the same controller; $K_{d,\psi} = -K_d$.

	error form, $K_d \dot{e}$	rate form, $-K_d r$
on a fixed command	identical	identical
at a step in ψ_d	349 N·m demanded	−1.1 N·m: nothing happens
at a $\pm 180^\circ$ seam crossing	12 566 N·m	nothing happens
sensor needed	a differentiated heading, and a filter	the gyro, as it is
when the command ramps	anticipates it	lags slightly behind
used in	this week's models, so that the kick can be measured	Week 5 , where guidance moves ψ_d at every waypoint

Experiment 4-3d · The kick, and what the rate form would have asked for (5 min)

What it measures. Lines 2 and 5: the D term actually demanded at the 10° step, and what $-K_d r$ would have demanded in the same run.

The run. The log of Experiment 4-3b, on `W04_E_PD` — no new model and no new simulation. Both forms are read from the same run, because both see the same vessel.

Opening and running.

```
W04_0_setup
R = W04_read('W04_E_PD', 'Kp', 300, 'Kd', 100);
r = gradient(deg2rad(R.psi), R.t);          % 기록된 선수각에서 회두율 / yaw rate from the log
[max(abs(R.D)) max(abs(Kd*r)) N_max]      % 오차 미분 / 회두율 되먹임 / 한계
```

Expected output:

```
ans =

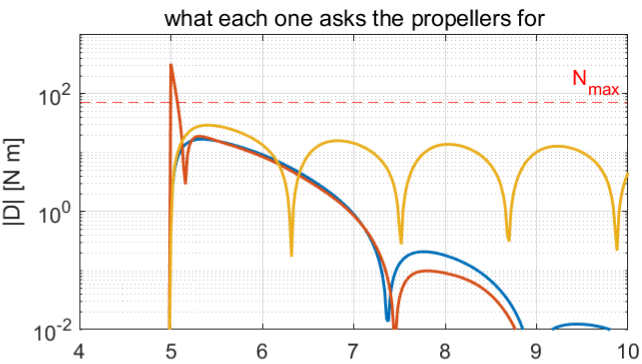
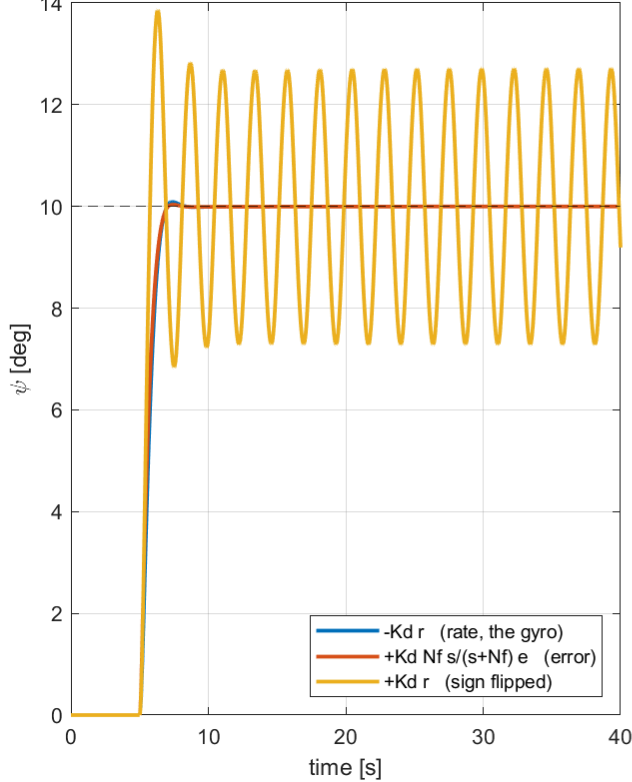
    325.7948    20.9746    70.8488
```

Reading the output.

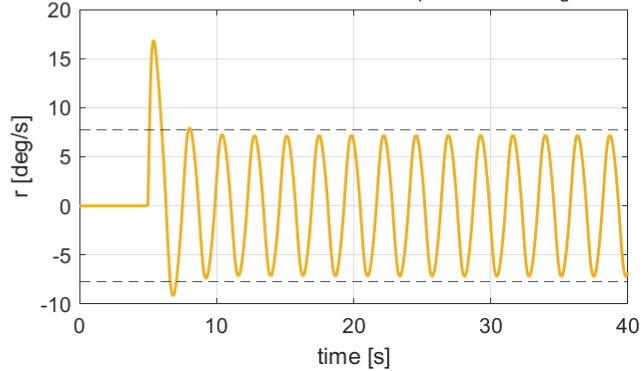
Printed	What it is	Which line it shows
325.79	the largest D term the models actually demand	line 2: the kick, 4.6 times the moment the propellers can produce
20.97	the largest $ K_d r $ in the same run	line 2: the rate form never asks for more than the vessel is doing — the peak yaw rate is 12.02 deg/s, and $100 \times 0.2097 = 21.0$ N·m
70.85	the moment limit	the line both are judged against

- At the instant of the step the two are **325.8** N·m and **−1.1** N·m. A second later, with the command no longer moving, they agree — which is line 1.

forms 1 and 2 lie on one another; form 3 is a brake wired backwards



form 3: a limit cycle at the rate where $|N_r|/(1+10|r|) = K_d$ (7.7 deg/s)



Reading the figure against the derivation.

Where to look	What is there	Which line predicts it
left panel, blue against red	one line, to the width of the stroke	the two forms are the same term while $\dot{\psi}_d = 0$
the first two rows of the table	0.90 against 0.39 % , 1.74 against 1.72 s	the same again, measured: what separates them is not the tracking
top right, the red spike at $t = 5$ s	above the dashed N_{max} , and off the top of the blue trace	line 2: 325.79 against 17.03 N·m, a factor of 19
top right, the blue trace at the same instant	no spike at all	line 2: r belongs to a vessel with inertia and cannot jump
left panel, the yellow trace	it reaches the command and then never stops crossing it	line 8: the bracket is negative, so every swing gains energy
bottom right, the yellow oscillation	held between the dashed lines at ± 7.70 deg/s	line 9: the hull's own nonlinear damping catches it there
the settle column, third row	never — it is still swinging at 40 s	line 8: nothing in the loop is removing energy at low rates

What the figure says

- Two of these three choices steer the vessel the same way and ask the propellers for numbers that differ by a factor of nineteen. The third is the same gain with one sign wrong, and it turns the brake into an accelerator.

What to try	What to watch
<pre>psi_step = 90; T_final = 80;</pre> then rerun	the two good forms are indistinguishable — overshoot 0.00 % , settling 5.58 and 5.60 s — while the demand parts by a factor of 88 : 33.27 against 2932.15 N·m
<pre>d_form = 3; Kd = 30;</pre> Run	$30 < N_r = 42.65$, so line 9's formula returns a negative rate — there is no crossing, the bracket is positive everywhere, and the vessel settles in 6.42 s with 18.45 % of overshoot. The wrong sign is survivable only while it is smaller than the damping it is fighting
<pre>d_form = 3; Kd = 200;</pre> Run	the limit cycle grows with the gain: 79.88 % of overshoot and a turn rate of 18.84 deg/s , against the $\frac{1}{10}(200/42.65 - 1) = 0.369$ rad/s = 21.14 deg/s line 9 predicts

In class

- **Purpose** — settle where the D term comes from, and why its sign is the whole of its meaning.
- **Point to** — the blue and red traces lying on one another, then the log axis beside them.
- **Ask** — "If the two track identically, why does the choice matter?" It is not the vessel that notices; it is the actuator, and the seam of §4-6.
- **Take away** — differentiate nothing that a sensor already measures, and check the sign of a damping term against the damping the hull already has.

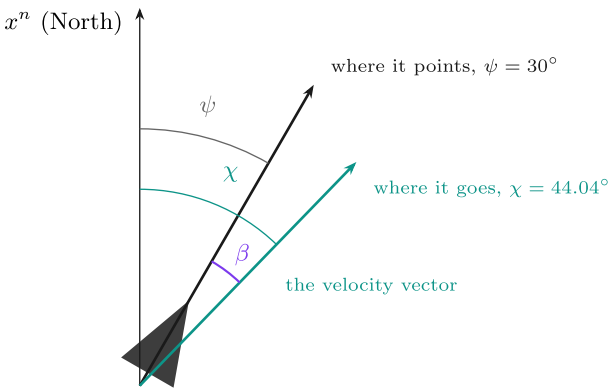
4-4. Heading is not course — the crab angle

This section answers: does a vessel go where it points?

- A heading controller regulates where the vessel **points**. It does not regulate where the vessel **goes**, and those are two different directions.

Heading, course and the crab angle

Where the vessel **points** and where it **goes** are two different directions. A heading controller regulates the first one only.



three angles, one identity

$\chi = \psi + \beta$

ψ heading — the hull's x axis, from North
 χ course over ground — the velocity, from North
 β crab angle — the gap between them, $\beta = \text{atan2}(v, u)$
 With $u = 2.0$ and $v = 0.5$: $\beta = 14.04^\circ$ and $\chi = 44.04^\circ$.

β is non-zero whenever the sway velocity v is — in a turn, in a current, in a beam wind. **It is not a disturbance to be rejected; it is what the hull is doing.** Steering ψ while travelling along χ is the permanent path error that Week 4 has to remove, and §4-7 measures it as $\Delta \tan \beta_c$.

In the figure	Meaning
orange ray	the heading ψ — the direction x_b points
blue ray	the course χ — the direction the velocity actually goes
green arc	the crab angle β , the gap between them

$\beta = \text{atan2}(v, u), \quad \chi = \psi + \beta$

Symbol	Definition	Unit
ψ	heading, from North to x_b	rad
χ	course over ground, from North to the velocity vector	rad
β	crab angle	rad
u, v	surge and sway velocity (Week 1)	m/s

- With the example of Week 1, $u = 2.0$ m/s and $v = 0.5$ m/s at $\psi = 30^\circ$: $\beta = \text{atan2}(0.5, 2.0) = 14.04^\circ$ and $\chi = 44.04^\circ$.
- The crab angle is non-zero whenever $v \neq 0$ — in a turn, in a current, in a beam wind. It is what the hull is doing, not an error to be removed. Week 1 measured $\beta = -20.3^\circ$ in a turning run of a vessel with no sway actuation at all.
- Fossen writes $\beta = \arcsin(v/U)$; for $u > 0$ it is the same number. The form $\text{atan2}(v, u)$ remains correct going astern.
- This week regulates ψ and lets χ fall where it may. **Week 5 cannot:** a path-following law that steers the heading while the vessel travels along the course leaves a cross-track error of the order of β times the look-ahead distance.

Experiment 4-4 · The crab angle, from two velocities (3 min)

What it measures. The formula above, on the numbers Week 1 measured. There is no model here: the crab angle is a property of the velocity, and two lines in the Command Window are the whole calculation.

Opening and running.

```

u = 2.0;  v = 0.5;  psi = 30;           % 1주차의 값 / the Week 1 example
beta = atan2d(v, u)                   % 크랩각 / the crab angle
chi = psi + beta                       % 침로 / the course over ground

```

Expected output:

```

beta =

    14.0362

chi =

    44.0362

```

What to try	What to watch
$v = -0.5;$	$\beta = -14.04^\circ$: sway to port carries the vessel to port of where it points
$v = 0;$	$\beta = 0$ and $\chi = \psi$ — the only case in which a heading controller is also a course controller
$u = -2.0; v = 0.5;$	$\beta = 165.96^\circ$: going astern, atan2 still returns the true direction of travel, where $\arcsin(v/U)$ would not

- The vessel of this week is never asked for a particular χ , so the crab angle is only observed. Week 5 steers a **path**, and then β becomes the standing error that ILOS has to remove.

4-5. The tuning order, applied to the heading

This section answers: how are the gains chosen for this axis, without a model — and what does the moment limit add?

The procedure of Week 2 §2-12. The requirement: a 10° turn inside 2 % within 3 s with overshoot below 5 %; no heading error with a weak propeller; no windup in a 90° turn.

Step	What is done	What was measured	Decision
0	requirement; is the plant well behaved?	§4-2: stable turn rate, integrating heading, mildly nonlinear	proceed
1	P only; scale of K_p from the units	a 10° error is 0.175 rad; $K_p = 300$ asks 52 N·m of the 70.85 available	$K_p = 300$
2	read the P response	overshoot 12.2 %, settling 3.26 s: it rings	add D
3	raise K_d while the response improves	$K_d = 25$: 7.61 %, 2.98 s; 50: 4.10 %, 2.22 s; 100: 0.39 %, 1.72 s; 150: 0 %, 2.42 s	$K_d = 100$, the fastest settling
4	add I if something pushes back	a 70 % port propeller leaves 0.80° ; $K_i = 20$ removes it, settling 1.98 s	$K_i = 20$
5	look at the moment: a big turn	a 90° turn saturates the moment; the back-calculation gain K_b decides what the integral does meanwhile	below

The moment limit and K_b . Measured by Experiment 4-5, below, settling time and overshoot for three turns:

K_b	10° turn	weak propeller, 10°	90° turn
0 (no anti-windup)	12.42 s, 4.35 %	1.98 s, 0 %	37.12 s, 15.57 %
0.05	7.28 s, 3.20 %	2.18 s, 0 %	24.14 s, 6.38 %
0.1	2.64 s, 2.06 %	9.38 s, 0 %	5.80 s, 0 %
1	33.68 s, 0 %	35.00 s or more, 0 %	48.50 s, 0 %

- **Too little K_b : windup.** While the moment is on its limit the integral keeps charging — to 79.7 N·m in the 90° turn — and the vessel overshoots by 15.6 % and takes 37 s to come back. Even the 10° turn winds up a little: the derivative kick at the step touches the limit for a moment.
- **Too much K_b : the integral is dragged down.** While saturated, back-calculation pulls the integral towards the value that makes the demand equal to the limit. With $K_p e$ alone asking for several hundred N·m, that value is far below zero — -249.5 N·m at $K_b = 1$ — and the vessel stalls short of its heading for tens of seconds. This is the I^* of Week 2 §2-11, on the vessel.
- $K_b = 0.1$ is the compromise: the 90° turn behaves as if there were no integral at all (5.80 s against 5.60 s for PD alone), the 10° turn meets the requirement, and the weak propeller is still corrected, more slowly.

The result, $K_p = 300$, $K_d = 100$, $K_i = 20$, $K_b = 0.1$, is the default in `w04_0_setup.m`. No transfer function of the vessel was written down.

Experiment 4-5 · The tuning order, and a big turn (15 min)

What it measures. The five steps of the table above, and then the one number that only a saturating turn can decide: K_b , measured on three turns at once.

The model. `w04_H_tuning` — the full loop with the moment limit and the back-calculation path. The 90° turn is what drives the moment onto its limit for long enough to matter.

Opening and running.

```

W04_0_setup
psi_step = 90; T_final = 60;
open_system('W04_H_tuning')      % Run - Kb = 0.1: PD 만큼 깔끔하다 / as clean as PD alone
Kb = 0;                          % Run - 적분이 쌓여 15.6 % 넘친다 / windup, 15.6 % overshoot
Kb = 1;                          % Run - 너무 끌어내려 한참 못 미친다 / dragged down, and slow
W04_H_tuning_by_hand            % 다섯 단계와 Kb 세 선회를 한 번에 / all five steps and three turns

```

Expected output:

W04 Experiment 4-5 the tuning order on the heading

1-2 $K_p = 300$ (10 deg asks 52 N m of the 70.8 available): overshoot 12.2 %, settle 3.26 s: it rings, so D

3 $K_d = 25$: 7.61 %, 2.98 s 50: 4.10 %, 2.22 s 100: 0.39 %, 1.72 s 150: 0.00 %, 2.42 s -> $K_d = 100$ (fastest settling)

4 weak port propeller (0.7): PD leaves 0.80 deg; $K_i = 20$ removes it: 0.007 deg left, settle 1.98 s

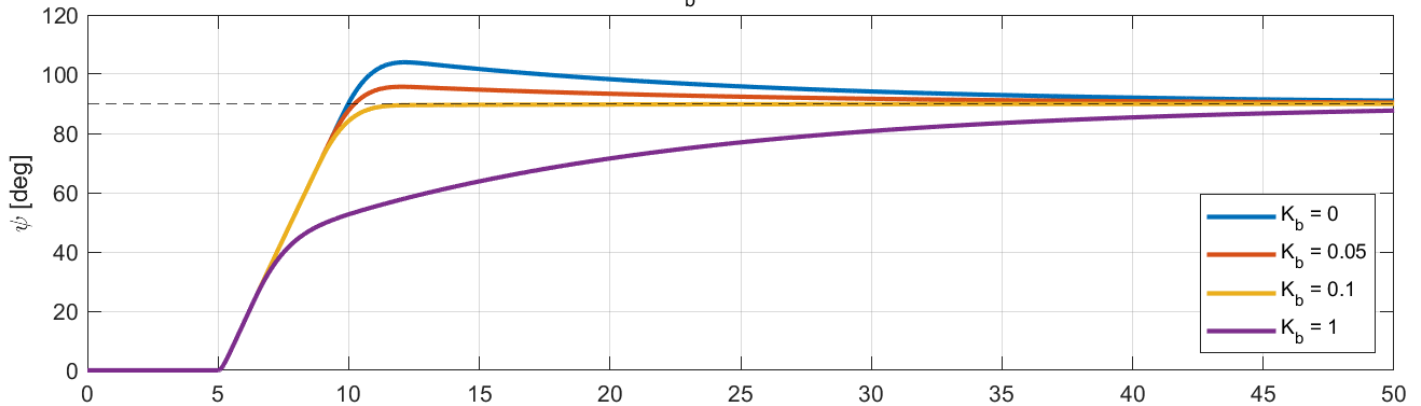
5 the moment limit: K_b against three turns (settle [s], overshoot %)

K_b	10 deg turn		weak propeller		90 deg turn	
0	12.42 s	4.35 %	1.98 s	0.00 %	37.12 s	15.57 %
0.05	7.28 s	3.20 %	2.18 s	0.00 %	24.14 s	6.38 %
0.1	2.64 s	2.06 %	9.38 s	0.00 %	5.80 s	0.00 %
1	33.68 s	0.00 %	35.00 s	0.00 %	48.50 s	0.00 %

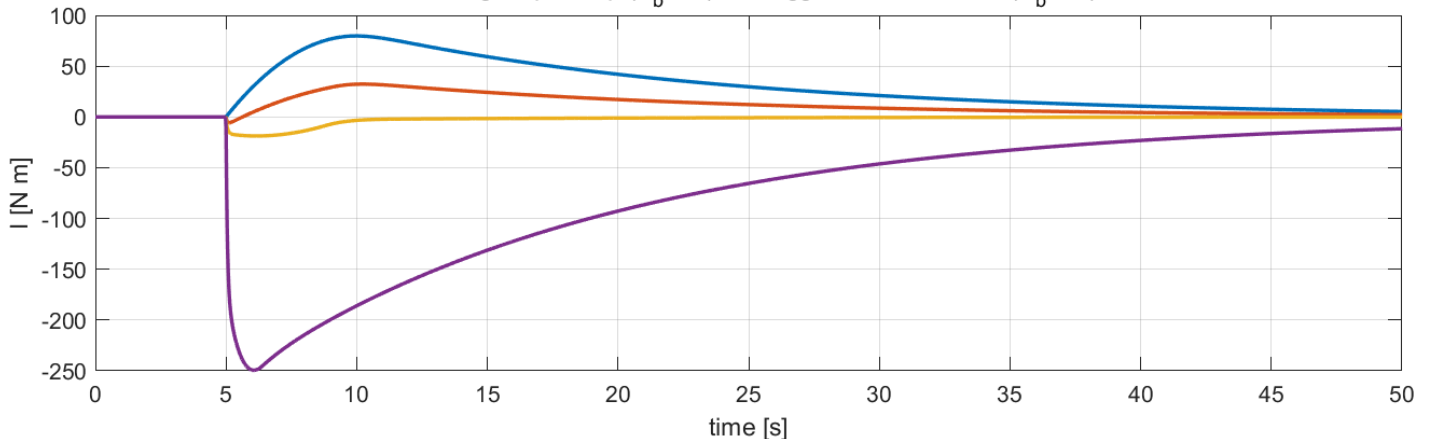
PD alone, 90 deg turn: 5.60 s, 0.00 %

final: $K_p = 300$, $K_d = 100$, $K_i = 20$, $K_b = 0.1$

a 90 deg turn: too little K_b overshoots, too much stalls short



the integral: piled up ($K_b = 0$) or dragged far below zero ($K_b = 1$)



In the figure	Meaning
top	the heading in a 90° turn for $K_b = 0, 0.05, 0.1, 1$
bottom	the integral term

Reading the figure against the steps.

Where to look	What is there	Which step or rule it shows
top panel, all four curves before about 8 s	one line	the moment is on its limit and the vessel turns as fast as it can, whatever the integral is doing
bottom panel, the $K_b = 0$ curve climbing	to 79.7 N·m	step 5, windup: the error is large for seconds, and nothing drains the store
top panel, the $K_b = 0$ curve past 90°	15.57 %, and 37 s to return	Week 2 §2-11: the store can be spent only by going past the command
bottom panel, the $K_b = 1$ curve diving	to -249.5 N·m	Week 2 §2-11's I^* : with $K_p e$ asking hundreds of N·m, the value that cancels it is far below zero
top panel, the $K_b = 1$ curve stalling short	tens of seconds	the same: that store has to be undone before the heading can finish
top panel, the $K_b = 0.1$ curve	5.80 s, no overshoot — PD alone gives 5.60 s	step 5's compromise: the integral is neither stored nor over-drained

What the figure says

- With too little K_b , the integral piles up and the vessel overshoots; with too much it is dragged far below zero and the vessel stalls short. $K_b = 0.1$ turns as cleanly as PD alone.

What to try	What to watch
<code>psi_step = 10; Kb = 0; Run</code>	even a small turn winds up a little: the moment touches its limit for an instant and the settling grows to 12.42 s
<code>Ki = 0; psi_step = 90; Run</code>	with no integral there is nothing to wind up, and K_b makes no difference at all — the remedy exists only because the term does

In class

- Purpose** — the full procedure, and one decision that cannot be made without looking at the actuator.
- Point to** — the blue integral rising to 80 N·m and the purple one diving to **-250** N·m, while the heading curves separate from about 8 s on.
- Ask** — "Why does a larger K_b make things worse here, when it helped in Week 3?" In a big turn $K_p e$ alone asks for several hundred N·m; back-calculation drags the integral towards the value that would cancel it, far below zero, and that has to be undone afterwards.
- Ask** — "Why is the weak-propeller case slower at $K_b = 0.1$?" The derivative kick at the step touches the limit, back-calculation pulls the integral down, and the integral must then climb to the **4.19** N·m the fault needs.
- Take away** — the back-calculation gain is a trade-off, and it is chosen by running the turns that matter.

4-6. The wrap

This section answers: why is the heading error passed through `ssa`?

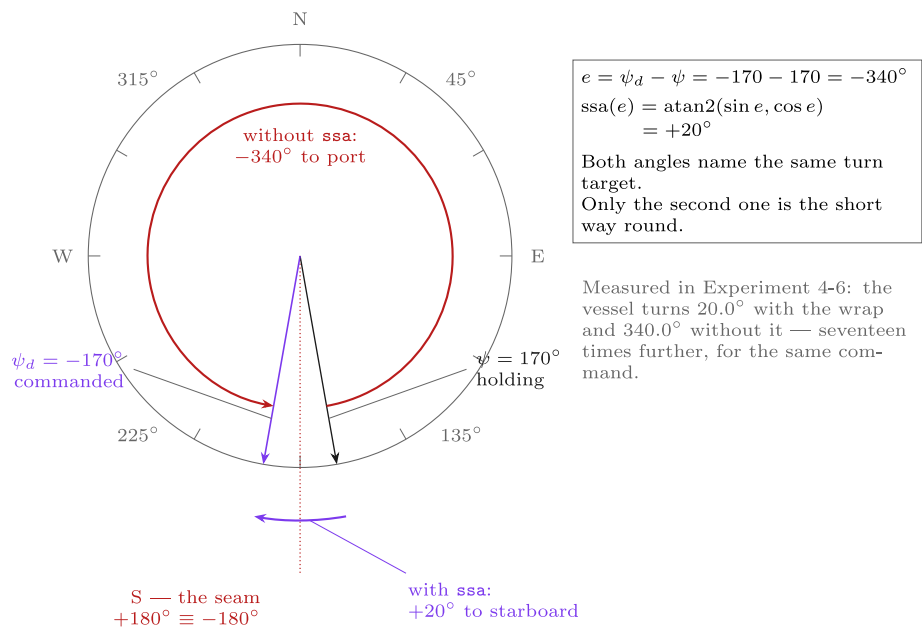
What is observed. The vessel holds 170° and is commanded to -170° . With one block removed from the error path it turns **all the way round the other way: 340°** instead of 20° .

What follows from it, one line at a time.

- 1. An angle and that angle plus 360° are the same heading, so the difference of two angles is only defined up to 360° . The raw subtraction $-170 - 170 = -340^\circ$ is arithmetically correct and physically absurd.
- 2. A controller acts on the number it is given: -340° asks for a 340° turn to port, and the vessel obeys.
- 3. The remedy is to map the error onto the shortest equivalent angle, $ssa(e) = atan2(\sin e, \cos e)$, which returns a value in $(-180^\circ, 180^\circ]$ — here $+20^\circ$. The sine and the cosine are unchanged by adding 360° , which is exactly why they are used.
- 4. This is not an edge case. Week 5's guidance produces commanded headings anywhere in $(-180^\circ, 180^\circ]$, and a mission of a few legs crosses the seam routinely.

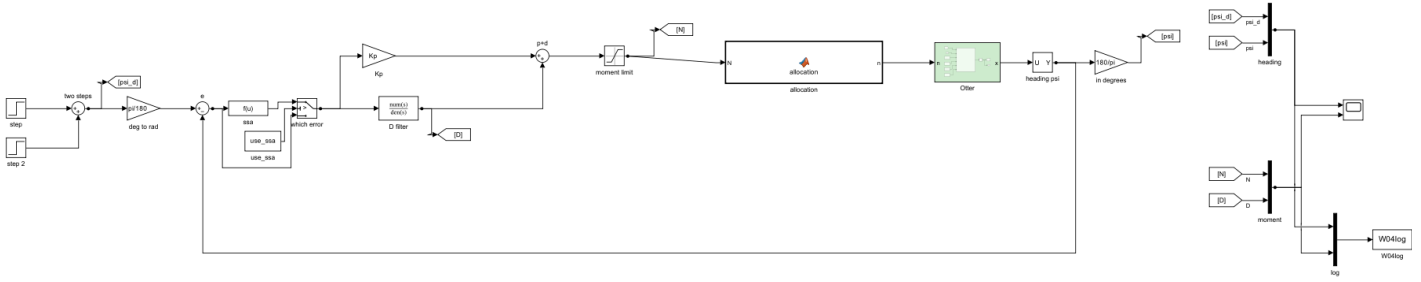
Why the heading error is wrapped

The vessel holds $\psi = 170^\circ$ and is commanded to $\psi_d = -170^\circ$. The two headings are 20° apart, on either side of the $\pm 180^\circ$ seam.



In the figure	Meaning
the grey circle	every heading, measured from North clockwise, as the vessel's compass reads it
the dotted red line at S	the seam, where $+180^\circ$ and -180° are the same direction and the number jumps
the black ray	where the vessel is pointing, $\psi = 170^\circ$
the violet ray	where it is asked to point, $\psi_d = -170^\circ$, which is 190°
the violet arc outside the circle	the turn the wrapped error asks for: $+20^\circ$ to starboard
the red arc inside the circle	the turn the raw subtraction asks for: -340° to port, the whole circle but for 20°
the box on the right	the two numbers, and the one line that turns the second into the first

Reading the figure, one angle at a time.



WEEK 4, SECTION G - THE WRAP AT ± 180 DEG

$e = \text{wrap}(\psi - \psi_d)$ in rad, $N = K_p e + K_D (d\psi/dt)$ in rad/s

N is limited to $\pm N_{\text{max}}$, what the proportional gain gives while putting K_D ahead

ψ_d step = 170, ψ_d step2 = -170, $t_{\text{step2}} = 20$. Then use $\text{use_ssa} = 1$ and Run again

Change a gain in the Command Window (e.g. $K_p = 100$) and press Run

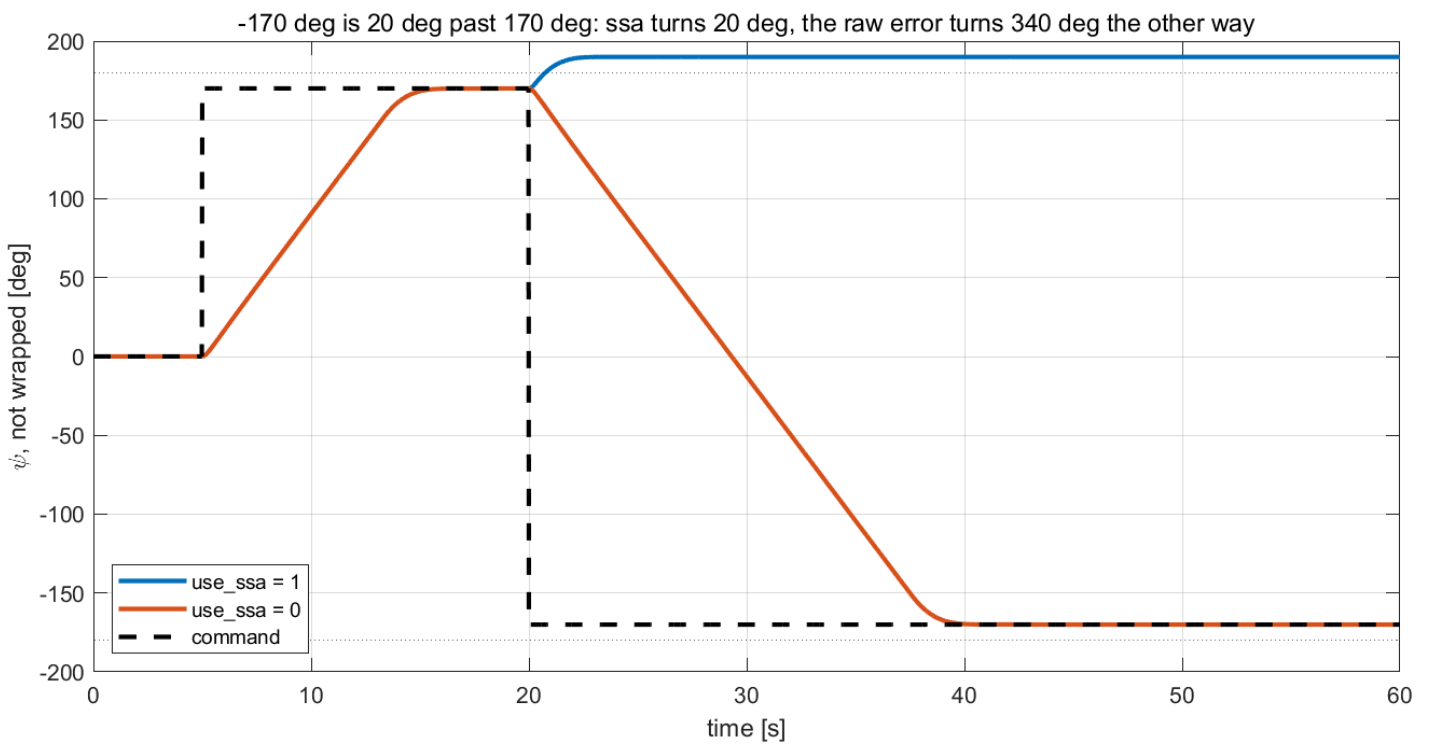
The Scope shows the heading in degrees (top) and the raw moment (bottom)

In the figure	Meaning
<code>ssa</code> , <code>use_ssa</code> , which error	the wrapped error when <code>use_ssa = 1</code> , the raw error when 0
<code>step</code> , <code>step 2</code> , <code>two steps</code>	170° at 5 s, then -170° at <code>t_step2</code>

Expected output:

W04 Experiment 4-6 170 deg, then -170 deg at $t = 20$ s

use_ssa	heading at 20 s [deg]	heading at 60 s [deg]	turned after 20 s [deg]
1	170.0	190.0	20.0
0	170.0	-170.0	340.0



In the figure	Meaning
traces	the heading, not wrapped, with and without <code>ssa</code> ; the dotted lines are $\pm 180^\circ$

Reading the figure against the derivation.

Where to look	What is there	Which line predicts it
both traces up to $t = 20$ s	identical, at 170°	the seam has not been crossed yet, and the two error paths agree everywhere else
the two traces leaving 170° in opposite directions	$+20^\circ$ and -340°	line 2: the controller turns the way the number it was handed points
where each trace ends	190° and -170°	line 1: the same heading, written two ways. The plot leaves the heading unwrapped so that the turn is visible
the 340° trace crossing the dotted $\pm 180^\circ$ lines	it sweeps the whole circle	line 3: what <code>ssa</code> exists to prevent

What the figure says

- 190° and -170° are the same heading. With `ssa` the vessel reaches it by a 20° turn; without it, by 340° the other way.

What to try	What to watch
<code>psi_step2 = 150;</code> with <code>use_ssa = 0</code>	no seam is crossed, and the two settings agree exactly: the bug hides until the command happens to cross $\pm 180^\circ$
<code>psi_step = -170;</code> <code>psi_step2 = 170;</code> with <code>use_ssa = 0</code>	the same fault in the other direction — a 340° turn to starboard

In class

- Purpose** — one line of code, and the cost of leaving it out.
- Point to** — the two curves leaving 170° in opposite directions at 20 s.
- Ask** — "Why does the plotted heading end at 190° and not at -170° ?" The plot shows the heading without wrapping, to make the direction of the turn visible; physically they are the same.
- Take away** — wrap every heading error before any gain sees it.

Part 2 • Laboratory run order

The experiments of Part 1 are worked through in order; this table is the index of what was run, for repeating the week at home.

Every example has **its own model**, laid out as in Weeks 2 and 3: the controller on the left, the thrust split and the vessel in the middle, **one Scope** on the right with the heading in degrees on top and the yaw moment below.

Experiment	Model	Script	What it shows
4-0	all of them	<code>W04_0_setup</code> , <code>W04_1_build_heading</code>	the parameters, and every model written from code
4-1	<code>W04_H_tuning</code>	—	the Week 3 canvas with <code>ssa</code> in front and the thrust split behind
4-2	<code>W04_C_open_loop</code>	<code>W04_C_plant_from_outside</code>	a step yaw moment, no controller
4-3a	<code>W04_D_P</code>	<code>W04_D_proportional_only</code>	P only
4-3b	<code>W04_E_PD</code>	<code>W04_E_derivative</code>	P + D
4-3c	<code>W04_F_PID</code>	<code>W04_F_weak_propeller</code>	P + I + D, with a weak port propeller
4-3d	<code>W04_E_PD</code> , reread	—	both D forms read out of one run: the kick, and what the rate form would have asked
4-3e	<code>W04_I_rate</code>	<code>W04_I_rate_feedback</code>	the three D terms, each one actually steering: the gyro, the error, and the wrong sign
4-4	none — two lines in the Command Window	—	the crab angle from u and v
4-5	<code>W04_H_tuning</code>	<code>W04_H_tuning_by_hand</code>	the tuning order, a big turn, and the back-calculation gain
4-6	<code>W04_G_wrap</code>	<code>W04_G_the_wrap</code>	a command across $\pm 180^\circ$, with an <code>ssa</code> switch

- The scripts only repeat what Run already shows, for several gains at once, and print the numbers quoted in Part 1.
- Each experiment ends with an **In class** note: purpose, what to point at, a question with its answer, and the sentence to take away.

Summary

Week Summary

Step	What was done	How it was verified
1	placed the Week 2 controller around the heading, with <code>ssa</code> and a thrust split	<code>check_overlaps</code> 0 in seven models
2	measured the yaw axis from outside	Experiment 4-2: the heading grows without end; turn rate within 0.36 s to 63 %; rate per N·m falls from 0.610 to 0.407
3	P alone	Experiment 4-3a: no error at any gain; overshoot 5.8 → 15.9 %
4	D damps	Experiment 4-3b: 12.18 → 0.39 %, fastest at $K_d = 100$
5	I against a weak propeller	Experiment 4-3c: 0.80 ° removed; integral 4.15 N·m
5a	settled where the D term comes from, and why its sign is negative	Experiment 4-3e: the rate and error forms track alike (0.90 against 0.39 %) and demand 17.03 against 325.79 N·m; the sign flipped never settles, limit-cycling at 7.23 deg/s against the 7.70 predicted
6	the wrap	Experiment 4-6: 20 ° with <code>ssa</code> , 340 ° without
7	the tuning order and K_b	Experiment 4-5: $K_p = 300$, $K_d = 100$, $K_i = 20$, $K_b = 0.1$; 90 ° turn in 5.80 s without overshoot

Progress Check

★ Minimum condition for following Week 5

Theory

- ☐ Able to explain from one open-loop test why P alone leaves no heading error.
- ☐ Able to state what P, I and D do on the heading and compare with Weeks 2 and 3.
- ☐ Able to compute the crab angle and course from u , v and ψ .
- ☐ Able to explain both failure modes of the back-calculation gain.

Laboratory

- ☐ `w04_1_build_heading` ran and every model reported `overlapping lines: 0`.
- ☐ `w04_check(1)`, `(2)` and `(3)` pass on a model built from `w04_P1_start`.

Recorded observations

- ☐ The heading error left by PD with a 70 % port propeller, and the moment the integral found.
- ☐ The **90**°-turn settling time at $K_b = 0$, **0.1** and **1**.

In-class laboratory — build the heading autopilot by hand

The second hour of the Week 4 session is spent building the autopilot in Simulink around a vessel whose allocation is given.

```
cd lectures/W04_simulink/problems
W04_P1_start           % creates W04_P1.slx – hull and allocation only
W04_check(1)           % run this whenever, as often as needed
```

	Problem	Time	The number it must reproduce
1	P only	20 min	no error; overshoot 5.8 % at $K_p = 100$, 12.2 % at 300
2	add the filtered derivative	20 min	overshoot 12.18, 4.10, 0.39 % at $K_d = 0, 50, 100$
3	the wrap	20 min	a 20° turn with ssa , 340° without

- The problem sheet is `W04_simulink/problems/README.md`, and reference answers are in `W04_simulink/solutions/`.

Assignment 4

- Due:** before the Week 5 session
- Submit:** the modified `W04_0_setup.m`, the numbers requested below, and two figures

① Requirements

Halve the surge force, $X_{ff} = 30$ N, in `W04_0_setup.m` and `W04_vars.m`, and tune the heading loop again by the tuning order of §4-5 with the same requirement.

② Verification — mandatory

★ **A claim is not a result. Numbers are required, with the tolerance band stated**

- The new $N_{\max}$, computed from §4-1 and read from `W04_0_setup`'s output.
- Section C with the new X_{ff} : the turn rate per N·m at 10 N·m.
- For each step of the tuning order, the measurement that decided it.
- The K_b table of §4-5 for the final gains.

③ Analysis (5–10 lines)

Explain why the available yaw moment depends on the surge force, and how that moved the gains.

Grading

Criterion	Weight
The model runs and produces the requested output	25%
Verification performed and numbers reported	40%
Correctness of the analysis	25%
Readability of the code	10%

Troubleshooting

Symptom	Cause	Fix
<code>w04_0_setup</code> stops in <code>mss_path</code>	the MSS toolbox is not at <code>Tools\MSS</code>	place MSS there; see the first page
the vessel stalls short of a big turn for tens of seconds	K_b too large	<code>Kb = 0.1</code>
a big turn overshoots and creeps back	$K_b = 0$: windup	<code>Kb = 0.1</code>
the vessel turns the long way round	<code>use_ssa = 0</code>	<code>use_ssa = 1</code>
a Scope does not match these notes	a changed variable is still in the workspace	run <code>w04_0_setup</code> again
<code>allocation</code> reports that <code>x_ff</code> or <code>k_pos</code> is undefined	<code>w04_0_setup</code> was not run, so the block's parameters have no values	run <code>w04_0_setup</code> ; the block reads <code>x_ff</code> , <code>y_pont</code> , <code>k_pos</code> , <code>k_neg</code> and <code>port_eff</code> from the workspace

References

Primary

- Fossen, T. I. *Handbook of Marine Craft Hydrodynamics and Motion Control*, 2nd ed., Wiley, 2021, §12.2 (PID control of marine craft) and §12.2.6 (anti-windup); the crab angle and course are defined with the kinematics of Ch. 2.
- Åström, K. J. and Hägglund, T. *Advanced PID Control*, ISA, 2006, Ch. 3.
- MSS toolbox, `Tools\MSS\VESSELS/otter.m` and `GNC/ssa.m`. The rate form of the D term is what the toolbox itself steers by — `VESSELS/SIMclarke83.m`, `VESSELS/SIMremus100.m` and `VESSELS/SIMrig.m`; not one MSS vessel differentiates the heading error.

In this course

- Week 2 §2-4, §2-6 to §2-8, §2-11 and §2-12; Week 3 §3-3 and §3-5.
- The previous, model-based version of this week — the Nomoto model, pole placement and the second-order formulas — is kept in `w04_simulink/_previous_version/` for reference.

Next Week

- Week 5 — Waypoint Following and LOS Guidance**
- A heading autopilot follows a commanded heading; guidance decides what heading to command. Week 5 builds the line-of-sight law that turns a list of waypoints into ψ_d , and meets the crab angle of §4-4 as a cross-track error.